# Affan Muhammad Shaikh

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## Education

**Ontario Tech University** Oshawa, ON

*Bachelor of Information Technology (Honours) - Networking and IT Security* *Sept. 2024 - Expected Apr. 2028*

- Coursework: Computer Security, Cryptography, Cloud Computing, Advanced Networking.

## Experience

**Co-Founder and Platform Developer** May 2026 - Present

*SSIK IT Consulting & Solutions - Student-Led Initiative* *Ontario, Canada*

- Co-founded a student-led IT and cybersecurity initiative with one partner and now define **service scopes**, **privacy research**, control-review methods, and client-ready recommendations.
- Developed a private internal review platform for tracking organizations, evidence, findings, and review states with **access controls**, outbound-request safeguards, **audit logs**, and verified backups.

**Head of Internal Technical Operations** Jan. 2026 - Present

*Arch Narrative - Developing Nonprofit* *Oshawa, ON*

- Built and currently manage internal systems supporting **280 volunteers** and **116 students** across **66 active mentor-student pairs**.
- Deployed centralized **Google Forms/Sheets** workflows for account requests and support tickets with Drive-linked evidence; administer role-based departmental inboxes and documented successor procedures.

## Projects

**OTNow Canvas Companion** | *JavaScript, Chrome MV3, Firefox WebExtensions, Canvas APIs*

- Designed and published a cross-browser Canvas companion on the **Chrome Web Store** and **Firefox Add-ons** with deadline grouping, moved-date detection, reminders, course links, and offline caching.
- Restricted Canvas access to **2** allowlisted read-only endpoints, stored coursework locally, and made anonymous aggregate reporting strictly opt-in.
- Opt-in reporting recorded **1,351 organized Canvas items** from **15 consenting installations** as of Nov. 4, 2026; identifiable coursework remains local.

**Secure File Transfer** | *Python, TLS, SHA-256, FastAPI*

- Built an authenticated **TLS** client/server and loopback-only browser workspace for certificate setup, account management, resumable uploads, recipient inboxes, and verified downloads.
- Implemented recipient-scoped storage, exact-offset resumption, two-sided **SHA-256** verification, tamper quarantine, throttling, and audit logging across upload and download paths.
- Benchmarked with **16 concurrent workers**, completing **2K** verified TLS transfers across **1K** unique files while returning all recipient inbox records within bounded protocol frames.

**Cisco Networking Labs** | *Cisco IOS, Packet Tracer, Wireshark*

- Designed and verified multi-device routed and switched topologies using **IPv4/IPv6**, **VLANs**, **802.1Q** trunks, inter-VLAN routing, DHCP, STP, NAT, and OSPF/EIGRP fundamentals.
- Isolated connectivity and routing faults with **Cisco IOS** operational state, routing tables, ping, traceroute, and **Wireshark** captures; revalidated reachability after each correction.

**P2P Messaging (Collaborative WIP)** | *Python, FastAPI, TCP/IP, ChaCha20-Poly1305*

- Co-developing a local encrypted messenger using **Ed25519/X25519** device identities and **ChaCha20-Poly1305** messages, with signed peer cards, fingerprint verification, and encrypted local history.
- Validated **10K** protocol operations in a controlled loopback benchmark: **5K** authenticated deliveries with matching encrypted acknowledgements and **5K** captured-envelope replays rejected.

## Technical Skills

**Networking:** IPv4/IPv6, subnetting, VLANs, 802.1Q, inter-VLAN routing, DHCP, DNS, NAT, STP, OSPF/EIGRP fundamentals, routing and switching troubleshooting
**Security:** TLS, SHA-256, file-integrity monitoring, access control, audit logging, input validation, threat modelling
**Languages:** Python, JavaScript, TypeScript, HTML/CSS
**Systems:** Linux, Windows Server, Cisco IOS
**Tools/Platforms:** Google Workspace Admin, Wireshark, Packet Tracer, Git/GitHub, WebExtensions, Canvas API